

Leveraged-ETF Rebalancing Reversal

Phase A1 results: a test of the hypothesis on free data (SEC EDGAR + Yahoo), July 2022 to September 2026

+9.7 bps

Top-decile overnight reversal
(beta-adjusted, before costs)

t = 1.72

Date-clustered t-stat
for that mean

−0.07

Net Sharpe of simulated
strategy (half-spread + 2
bps)

171%

Share of gross P&L from
the 10 best days

Recommendation: stop. Don't pay for data. Leveraged-ETF rebalancing flow shows at most a small, fragile overnight reversal in single stocks. It falls short of the 10–20 bps bar, it isn't clearly separate from ordinary short-term reversal, and its P&L depends on a handful of days and names. The one reason for caution is that the effect appears only in 2026, the year leveraged-ETF flow peaked. That is worth watching cheaply, not funding.

Scorecard against the continue/kill criteria

Criterion (from the handoff)	What the data show	Result
Top bucket $\geq$ 10–20 bps beta-adjusted reversal before costs	Top decile of FlowRatio : +9.7 bps overnight (t = 1.72; 1.68 clustered by date and stock)	FAIL
Significant beyond generic short-term reversal	Flow coefficient t = −2.04 with date fixed effects, but t = −1.62 once lev-ETF stocks may reverse differently; matched-placebo gap +15.0 bps (t = 2.19)	MARGINAL
Not concentrated in a few days or names	Gross P&L is negative without the 10 best days (171% share) and without the top 5 names (131%): IONQ, ...	FAIL
Present in the most recent year	2026: +14.0 bps (t = 1.90); +22.5 bps vs matched stocks (t = 2.56)	PASS

What was tested

- **Universe:** 408 daily-reset single-stock leveraged/inverse ETFs on 237 US-listed stocks, including **116 funds that have since closed**. The source is SEC fund registrations 2022–2026, with each fund's underlying checked against the swaps it reports to the SEC.
- **Flow:** each fund's required rebalance trade, $T = A \cdot L \cdot (L - 1) \cdot r$, summed per stock and divided by an estimate of closing-auction size (10% of 20-day average dollar volume) to give **FlowRatio**.
- **Outcome:** close-to-next-open return, beta-hedged, signed against the flow. Positive means the price reverted, which is what the trade would earn.
- **Sample:** 53,526 stock-days with leveraged ETFs, plus 1,492,148 stock-days of S&P 1500 stocks as the comparison group. Earnings days are excluded.

1. The trade would have lost money for three years, then made it back in 2026

The simulation fades the flow each day in the stocks whose |FlowRatio| is in the top 10% of the trailing year, so there is no look-ahead in the cutoff. Rules: enter at the close, beta-hedge, exit at the next open. Size is 10% of predicted flow, capped at \$500k per name and \$5M gross.

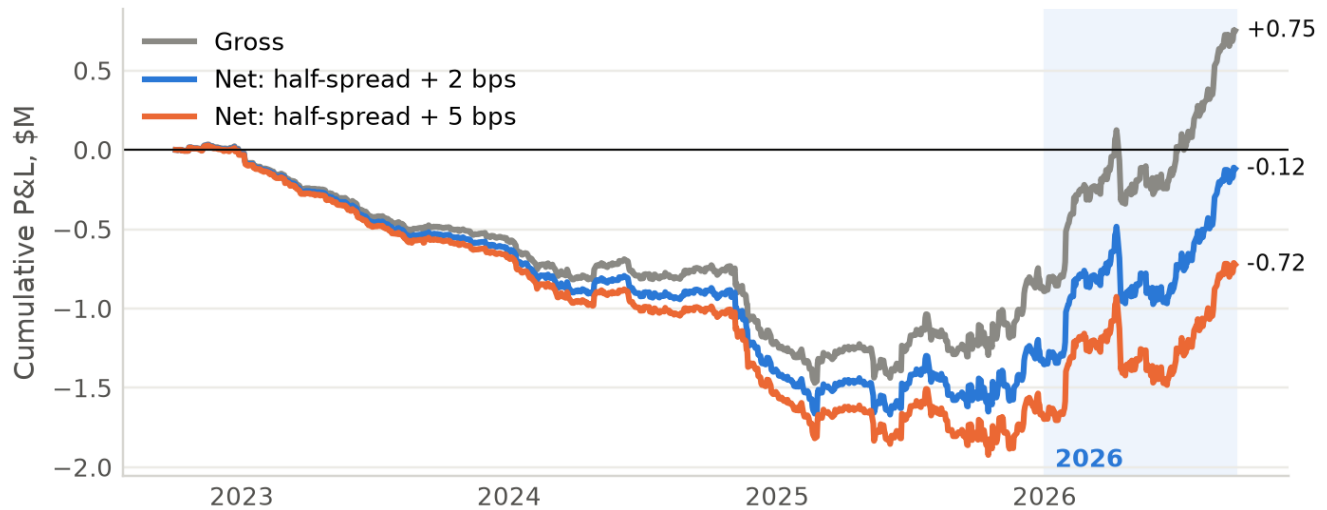


Figure 1. Cumulative P&L on \$5M of capital. Shaded: 2026. Costs are an assumed half-spread by liquidity bucket (1-20 bps) plus 2 or 5 bps, plus 0.5 bp for the hedge.

	Annual return	Volatility	Sharpe	Max drawdown	Total P&L
Gross	+3.6%	8.4%	+0.43	-30%	+\$0.75M
Net: half-spread + 2 bps	-0.6%	8.4%	-0.07	-34%	-\$0.12M
Net: half-spread + 5 bps	-3.5%	8.4%	-0.41	-39%	-\$0.72M
Net: full spread + 5 bps	-4.9%	8.4%	-0.58	-41%	-\$1.02M

The size-weighted gross edge is **3.7 bps per trade**, against about 4.9 bps of assumed cost. Average capital deployed was only \$2.2M of the \$5M.

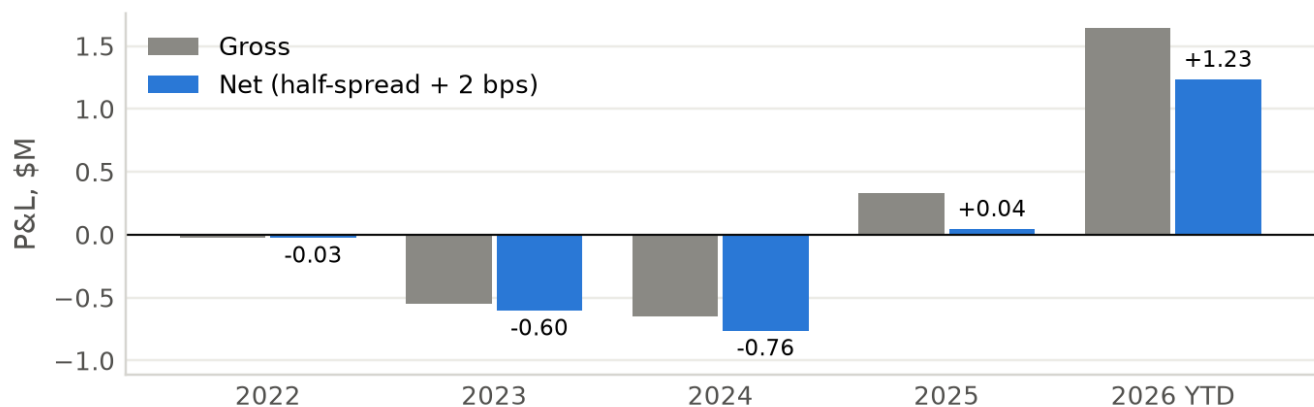


Figure 2. P&L by calendar year, \$M. All of the positive P&L comes from late 2025 and 2026.

2. Only the very largest flows reverse, and not by much

If rebalancing flow pushes the close away from fair value, reversal should rise steadily with flow size. It doesn't. Deciles 4–9 drift slightly with the flow, and only the top decile reverses. Even there the average is +9.7 bps with a 95% interval that includes zero.

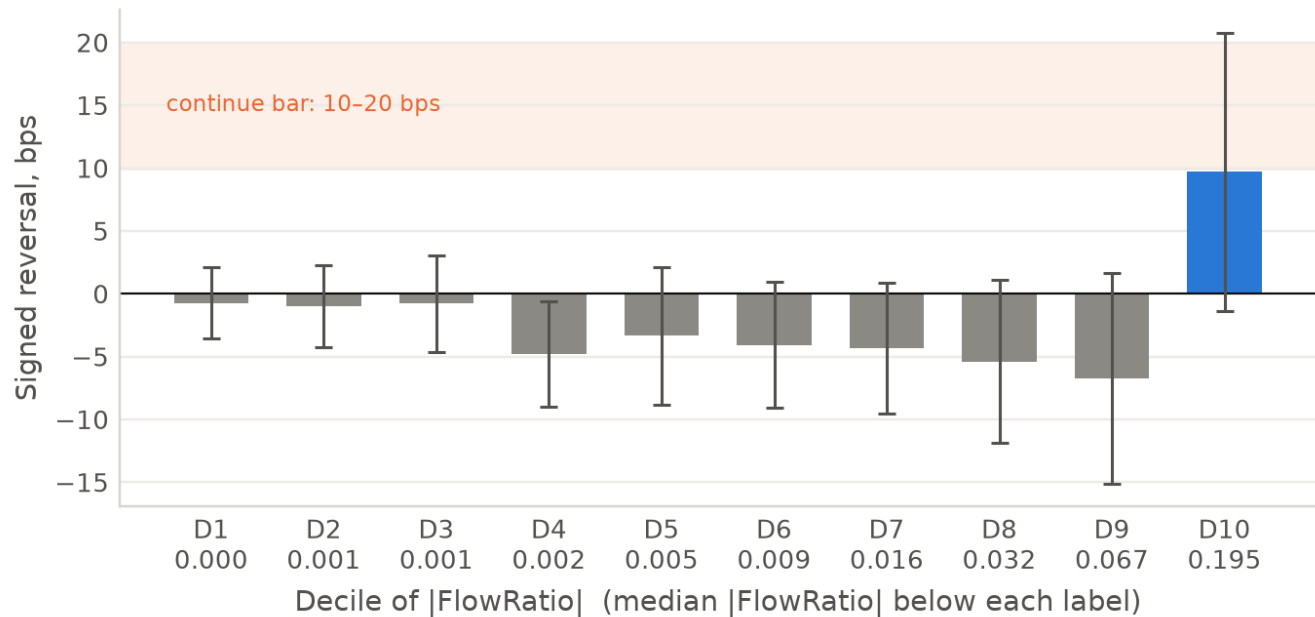


Figure 3. Overnight beta-adjusted signed reversal by decile of |FlowRatio| among stock-days with leveraged ETFs. Bars: mean, bps. Whiskers: 95% CI clustered by date. Shaded band: the 10–20 bps threshold.

Decile	Median FlowRatio	Median flow	Overnight, bps (t)	Close→close, bps (t)	Hit rate	Stocks
D1	0.000	\$0.0M	−0.7 (−0.5)	+0.2 (0.1)	48%	205
D2	0.001	\$0.1M	−1.0 (−0.6)	−3.8 (−1.1)	50%	215
D3	0.001	\$0.1M	−0.8 (−0.4)	+0.3 (0.1)	49%	223
D4	0.002	\$0.3M	−4.8 (−2.2)	−5.0 (−1.1)	50%	225
D5	0.005	\$0.5M	−3.3 (−1.2)	+2.9 (0.5)	49%	214
D6	0.009	\$1.0M	−4.1 (−1.6)	+0.1 (0.0)	49%	200
D7	0.016	\$1.6M	−4.3 (−1.6)	+3.3 (0.5)	49%	177
D8	0.032	\$2.8M	−5.4 (−1.6)	+5.8 (0.8)	49%	156
D9	0.067	\$5.7M	−6.7 (−1.6)	+3.5 (0.4)	49%	128
D10	0.195	\$26.7M	+9.7 (1.7)	+22.1 (1.7)	53%	102

Using fixed thresholds instead of deciles, flow between 30% and 100% of the auction proxy shows +28.7 bps ($t = 2.77$, 1,328 stock-days, 55 stocks). Flow above 100% turns negative again. This is one of six bins tested, so treat it as a nominal result rather than a finding.

3. Most of it looks like ordinary short-term reversal

Stocks with big daily moves tend to give some back overnight whether or not leveraged ETFs trade them. Two tests ask whether leveraged-ETF flow adds anything on top of that.

Same-size moves, with and without leveraged ETFs

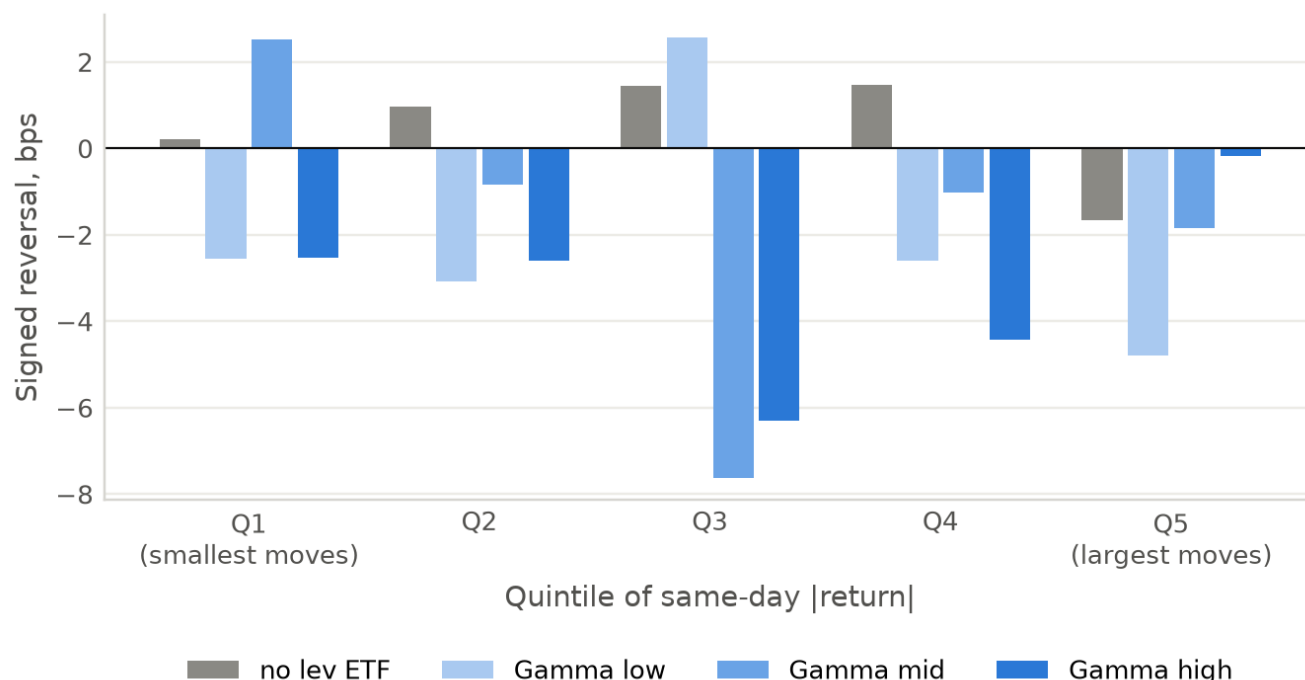


Figure 4. Within each quintile of same-day |return|: S&P 1500 stocks with no leveraged ETF (gray) vs leveraged-ETF stocks split into terciles of Gamma relative to liquidity (blue). More Gamma does not bring more reversal.

Regression with controls for the size of the move

Next-day overnight return regressed on the day's return, its absolute value, the prior day's return, return $\times$ liquidity, return $\times$ volatility and FlowRatio, with date fixed effects. The flow coefficient is negative as the hypothesis predicts. It is only borderline significant (-2.04) and weakens to -1.62 once leveraged-ETF stocks are allowed their own reversal slope.

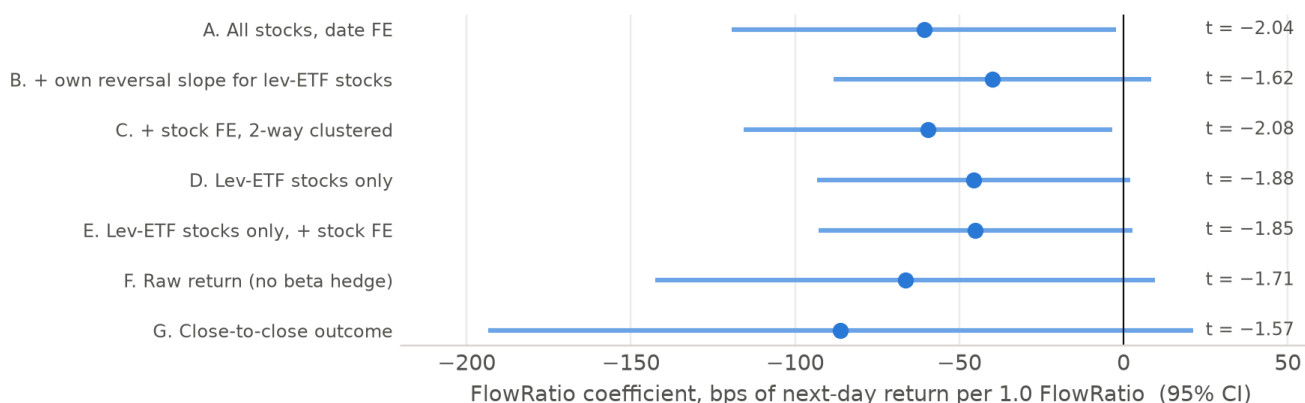


Figure 5. FlowRatio coefficient across specifications (95% CI). At the top-decile median FlowRatio of 0.20, a coefficient of about -60 means roughly 12 bps of reversal.

4. Against matched stocks, the gap is real only in 2026

Each top-decile stock-day is paired with the most similar stock that has never had a leveraged ETF: same day, same direction of move, closest in move size, dollar volume and volatility. Overall the leveraged-ETF stock reverses +9.7 bps and its match –5.2 bps, a gap of **+15.0 bps (t = 2.19)**. By year, the gap comes entirely from 2026.

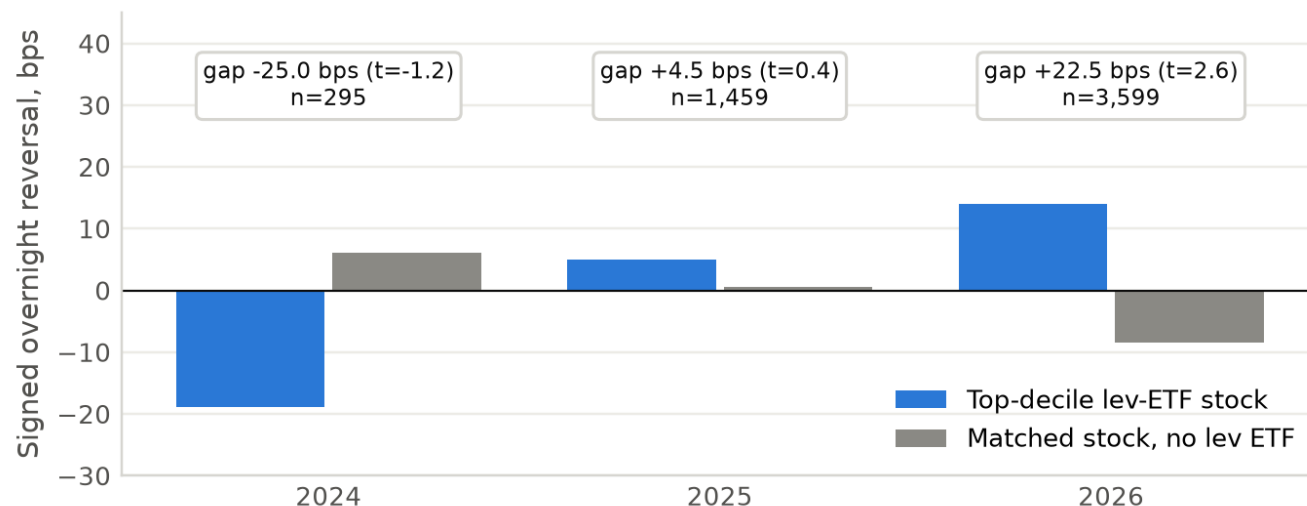


Figure 6. Top-decile stock-days vs matched non-ETF stocks, by year. The top decile has no 2022–23 days because flows were still small.

Splits of the top decile

Split	Group	Stock-days	Overnight, bps (t)	vs matched, bps (t)
Year	2024	295	–18.9 (–1.4)	–25.0 (–1.2)
Year	2025	1,459	+5.1 (0.5)	+4.5 (0.4)
Year	2026	3,599	+14.0 (1.9)	+22.5 (2.6)
Long vs inverse	mostly long (<20% inv Gamma)	4,589	+8.6 (1.6)	+14.9 (2.2)
Long vs inverse	mixed	526	+14.5 (1.2)	+20.0 (1.3)
Long vs inverse	inverse-heavy (>50%)	238	+20.7 (0.9)	+5.6 (0.2)
Liquidity	ADV low	1,785	+4.8 (0.5)	+12.4 (1.1)
Liquidity	ADV mid	1,784	+19.9 (2.8)	+23.2 (2.4)
Liquidity	ADV high	1,784	+4.4 (0.7)	+9.3 (1.1)
Flow direction	buy flow (up day)	2,635	+5.9 (0.7)	+6.7 (0.7)
Flow direction	sell flow (down day)	2,718	+13.5 (1.6)	+22.9 (2.4)
Options expiry	monthly opex	160	–1.0 (–0.0)	+53.4 (0.9)
Options expiry	non-opex	5,074	+8.6 (1.5)	+9.5 (1.5)
Options expiry	quad witching	119	+73.5 (1.6)	+194.9 (2.2)

No split is stable enough to build a rule on. The quad-witching figure rests on 119 observations, and with about 30 splits tested, one or two will look significant by chance.

5. Is the data good enough to trust a null?

A weak result is only informative if the flow estimates are right. The hard part is each fund's daily assets. They are rebuilt from SEC N-PORT filings (quarter-end net assets plus monthly creations and redemptions), each fund's own daily return, and a current-AUM anchor.

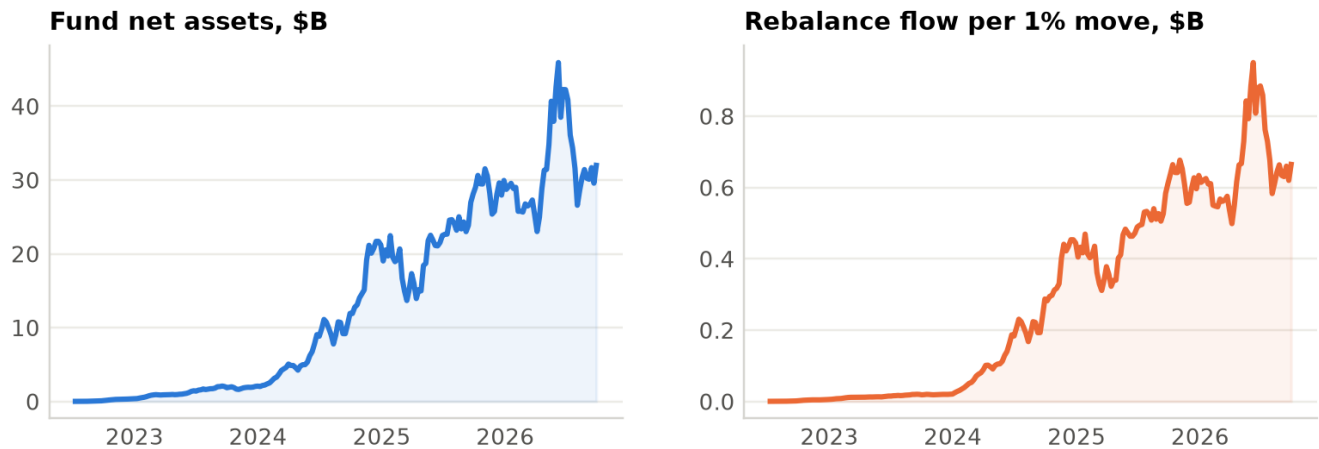


Figure 7. Reconstructed single-stock leveraged-ETF assets and the aggregate flow they generate for a 1% move in every underlying. Yearly average flow per 1%: 2024: \$0.19B, 2025: \$0.49B, 2026: \$0.67B.

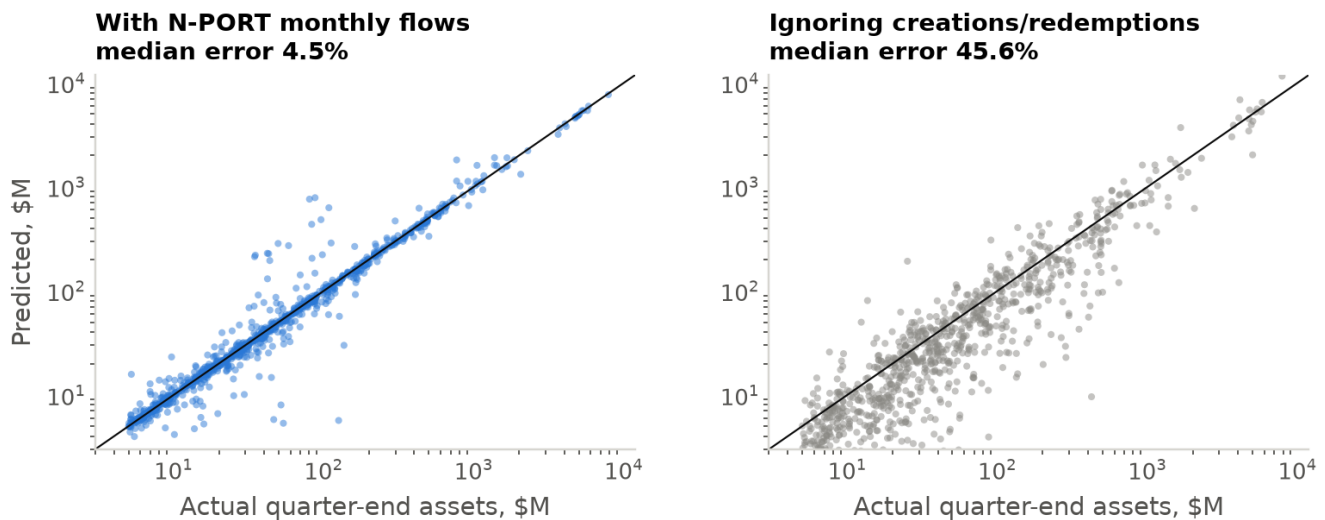


Figure 8. Hold-out test: predict each quarter-end N-PORT figure from the previous one (939 fund-quarters above \$5M). Using the monthly flow data cuts the median error from 46% to 4.5%.

- **External check:** the model gives **−\$2.4B** of MSTR rebalancing flow on 2024-11-21, consistent with press reports of \$2B+ single-day MSTR flows that month.
- **Leverage changes are handled.** Several funds ran at 1.5x before moving to 2x (TSLL, NVDL, CONL, ...). Leverage comes from each fund's measured daily returns, quarter by quarter.
- **Channel size:** reconstructed assets reach about \$33-40B in 2026, against the ~\$41-42B reported. The gap is crypto, private-company and non-US underlyings, which are excluded.
- **SEC filing errors are caught.** MSTZ's Nov-2024 report was also filed under a second series; the duplicate is removed.

6. Caveats, secondary check, and what I'd do next

What could still be wrong

- **How much flow actually hits the close is unknown.** Swap dealers can hedge during the day, and the 10%-of-volume auction proxy is a guess. If most flow is absorbed before 3:50 pm, a weak overnight reversal is what you'd expect. Only closing-imbalance data (Phase A2) settles this.
- **Mild look-ahead that favors the signal:** flow uses the full-day return (live, you'd estimate it at about 3:50 pm), and asset paths interpolate toward the next quarter-end filing. Neither can explain a weak result.
- **Costs are assumed.** There is no free quote history. Auction-to-auction trades avoid the spread but face adverse selection, which is not modeled.
- **Survivorship:** the 116 closed funds are included, but the comparison stocks are today's S&P 1500. Market cap is proxied by dollar volume, and Yahoo's open approximates the opening cross.

Secondary check: index-level leveraged funds

Index proxy	Funds	Avg daily flow	Flow → next overnight, bps per SD (t)
QQQ	PSQ, QID, QLD, SQQQ, TQQQ	\$1,808M	−2.6 (−0.87)
SPY	SDS, SH, SPDN, SPUU, SPXL...	\$575M	−0.4 (−0.15)
SOXX	SOXL, SOXS, SSG, USD	\$1,392M	−3.0 (−0.92)
XLK	REW, ROM, TECL, TECS	\$251M	−1.8 (−0.60)
IWM	RWM, SRTY, TNA, TWM, TZA...	\$208M	+1.8 (0.48)
XLF	FAS, FAZ, SEF, SKF, UYG	\$125M	+4.8 (1.48)
DIA	DDM, DOG, DXD, SDOW, UDOW	\$65M	+2.7 (0.98)
FXI	YANG, YINN	\$98M	−10.3 (−2.03)
XBI	LABD, LABU	\$96M	−1.5 (−0.55)
KRE	DPST, SKRE	\$44M	+0.5 (0.10)
XLE	DIG, DUG, ERX, ERY	\$14M	+7.1 (1.69)

Newey-West standard errors, 1,058 days. No index shows a reliable reversal after large leveraged-ETF flow; the one |t| ≈ 2 (FXI) is about what chance gives across 11 tests.

Next steps

Stop here and buy no data. Because the effect exists only in 2026, the year flow peaked, there is a cheap way to keep the option open. Run the zero-cost Phase A2 recorders (IBKR closing-imbalance logger, daily issuer holdings scraper) for one quarter, then re-test out of sample. That would also answer the key open question: how much of the predicted flow actually shows up in the closing auction.

Everything in this document is reproducible from the repository: universe in **universe/**, pipeline in **flows/**, statistics in **analysis/run_a1.py**, and all tables as CSV in **reports/**.